

Grust

covers grust (0.9.0)

A Rust Property Graph Architecture

Alexy Khrabrov

chiefscientist.org

&

Codex with ChatGPT 5.5

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Preface

Grust is a small Rust property graph API with a large architectural promise: application code should be able to build, validate, traverse, and persist graph data without committing itself to a database query language too early.

That goal matters because many graph-shaped Rust applications live between worlds. A crawler wants a local graph while it extracts facts. An indexing pipeline wants deterministic target state. A data product may begin with tests and an in-memory backend, then later need SurrealDB, FalkorDB, PostgreSQL, LanceDB, Sail, or another backend. Grust gives those applications one model to program against:

```
Graph = nodes + edges
Node  = id + label + properties
Edge  = optional id + from + to + label + properties
```

This book is a guided tour of that model. It covers the workspace architecture, the Rust concepts the code leans on, the core graph types, traversal IR, backend contracts, implemented backends, proposed backends, and practical examples.

1. The Shape of Grust

Grust is not trying to be every graph library for Rust. It is not an algorithm crate like `petgraph`, and it is not a thin wrapper around a single database. It is a property graph layer: a compact model of labeled nodes, labeled edges, and typed properties that can be carried across storage engines.

The current workspace is arranged around one core crate, one public facade, and backend or integration crates:

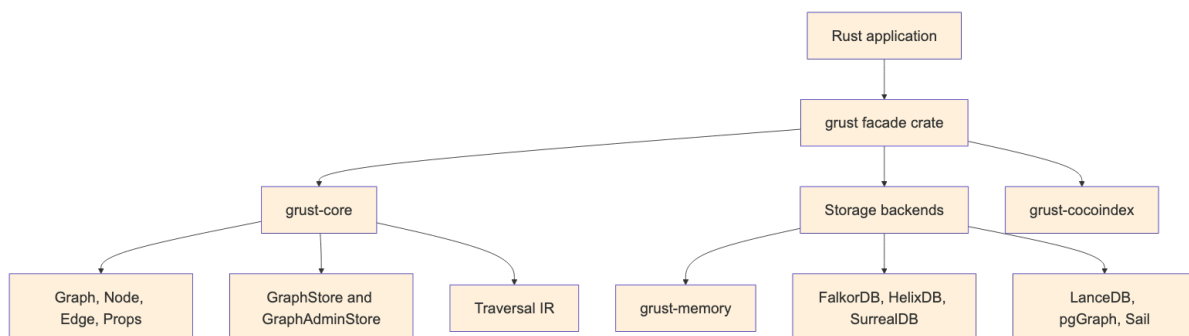


Figure 1: Diagram 1

The `grust-core` crate owns the durable concepts. It defines identifiers, labels, property values, graph builders, schemas, traversals, error types, load reports, and backend traits. The `grust` crate re-exports those pieces and gates backend crates behind Cargo features.

Core also owns the small lowering helpers that have to mean the same thing everywhere. `relationship_type` normalizes edge labels for backends that need database-safe relationship names. `schema_identifier` normalizes schema labels and fields for SQL-like typed surfaces. `edge_key` gives table and export backends the same structural fallback key for an edge when the caller has not provided an explicit `EdgeId`.

That split is important. Core model code stays light, deterministic, and mostly dependency-free. Backends are allowed to depend on Redis, SurrealDB, LanceDB, PostgreSQL, Spark Connect, HTTP clients, gRPC clients, or SDKs without dragging those dependencies into every Grust user.

2. The Core Property Graph

At the center of Grust are three public data structures:

```
pub struct Graph {
    pub nodes: Vec<Node>,
    pub edges: Vec<Edge>,
}

pub struct Node {
    pub id: NodeId,
    pub label: Label,
    pub props: Props,
}

pub struct Edge {
    pub id: Option<EdgeId>,
    pub from: NodeId,
    pub to: NodeId,
    pub label: Label,
    pub props: Props,
}
```

NodeId, EdgeId, and Label are newtypes over String. They are small wrappers, but they carry meaning through Rust's type system. A function that expects a Label cannot accidentally receive an EdgeId without an explicit conversion. The wrappers implement common conversion traits such as From<&str> and From<String>, so the public API remains ergonomic.

Properties are stored in a BTreeMap<String, Value>:

```
pub enum Value {
    Null,
    Bool(bool),
    Int(i64),
    Float(f64),
    String(String),
    StringArray(Vec<String>),
    Json(serde_json::Value),
}
```

The use of BTreeMap is a quiet but useful choice. Iteration order is stable, which makes generated JSON, tests, and backend query generation more predictable. That is especially helpful in a project whose backends turn one graph into SQL, Cypher-like queries, Arrow batches, JSON target state, or SDK calls.

Node::new also inserts an id property if one is not already present. That means a Grust node has a first-class typed identity and an ordinary property view of that identity, which helps backends that identify records through property maps.

3. Building Graphs

Most application code should not construct every Node and Edge by hand. It should use GraphBuilder:

```
use grust::prelude::*;

let mut builder = GraphBuilder::new();
```

```

let talk = builder
  .node("Talk", "talk:rust-graph-api")
  .prop("title", "A Modern Graph API for Rust")
  .prop("year", 2026_i64)
  .finish();

let speaker = builder
  .node("Person", "person:ada")
  .prop("name", "Ada Example")
  .finish();

builder
  .edge("PRESENTED_BY", &talk, &speaker)
  .prop("source", "conference-schedule")
  .finish();

let graph = builder.build();

```

The builder deduplicates nodes by `NodeId`. If an existing node has the same label, additional properties are merged into it. Edges default to `EdgePolicy::DedupeByFromLabelTo`, so the tuple (from, label, to) is treated as the relationship identity. Domains that need true multi-edges can opt into `EdgePolicy::AllowDuplicates`.

```

let mut builder = GraphBuilder::new()
  .edge_policy(EdgePolicy::AllowDuplicates);

```

This is an example of Grust's overall style: put the common property-graph case on the simple path, but leave an explicit escape hatch for graph models with different identity rules.

Loading and Saving Graph Documents

A Graph can also move through textual document formats without touching a backend. The core crate exposes paired constructors and serializers for YAML, JSON, and XML:

```

let graph = Graph::from_yaml(yaml_text)?;
let yaml_text = graph.to_yaml()?;

let graph = Graph::from_json(json_text)?;
let json_text = graph.to_json()?;

let graph = Graph::from_xml(xml_text)?;
let xml_text = graph.to_xml()?;

```

These methods are useful for fixtures, migration inputs, examples, audits, and small graph interchange files. They all feed the same validation path, so a document with duplicate node ids or an edge that points at a missing node fails before it reaches a store.

YAML and JSON share the same document shape. A property value can be written as a plain JSON-like scalar when the type is obvious, or as the tagged Grust Value representation when the exact variant matters:

```

{
  "nodes": [
    {
      "id": "talk:rust-graph-api",
      "label": "Talk",
      "props": {
        "title": "A Modern Graph API for Rust",

```

```

    "year": 2026,
    "tracks": {
      "type": "string_array",
      "value": ["rust", "graphs"]
    }
  },
  {
    "id": "person:ada",
    "label": "Person",
    "props": {
      "name": "Ada Example"
    }
  }
],
"edges": [
  {
    "label": "PRESENTED_BY",
    "from": "talk:rust-graph-api",
    "to": "person:ada",
    "props": {
      "source": "conference-schedule"
    }
  }
]
}

```

The XML form is more explicit because XML has no native object or array type. Properties are represented as repeated prop entries with a key and a tagged value:

```

<graph>
  <nodes>
    <node>
      <id>talk:rust-graph-api</id>
      <label>Talk</label>
      <props>
        <prop>
          <key>title</key>
          <value>
            <type>string</type>
            <value>A Modern Graph API for Rust</value>
          </value>
        </prop>
      </props>
    </node>
    <node>
      <id>person:ada</id>
      <label>Person</label>
    </node>
  </nodes>
  <edges>
    <edge>
      <label>PRESENTED_BY</label>
      <from>talk:rust-graph-api</from>
      <to>person:ada</to>
    </edge>
  </edges>
</graph>

```

```
</edges>
</graph>
```

Serialization removes the generated `id` property from node property maps when it only mirrors the node's `NodeId`. That keeps exported documents readable without changing what `Node::new` guarantees after loading.

4. Typed Graph Ingestion with `garde` and `zod-rs`

The core `Graph`, `Node`, and `Edge` types are intentionally dynamic. A node has a label and a property map; an edge has a label, endpoints, and a property map. That dynamic shape is the right interchange format for backends, graph documents, audits, and cross-system movement. Application code, however, often starts with typed Rust structs: a `Person`, a `Project`, a `WorksOn` relationship, or some domain-specific event. Grust's typed ingestion layer connects those two worlds.

The typed layer is optional. It is enabled through Cargo features:

```
[dependencies]
grust = { package = "grust-graph", version = "0.9.0", features = ["typed-garde"] }
```

`typed-garde` adds Rust-struct validation and typed lowering. A second feature, `typed-zod-rs`, layers raw JSON shape validation on top:

```
[dependencies]
grust = { package = "grust-graph", version = "0.9.0", features = ["typed-zod-rs"] }
```

`typed-zod-rs` implies `typed-garde`. That relationship matters: `zod-rs` checks untrusted JSON shape first, Serde turns the JSON into Rust values, and `garde` checks typed domain invariants before Grust lowers the value into a normal `Node` or `Edge`.

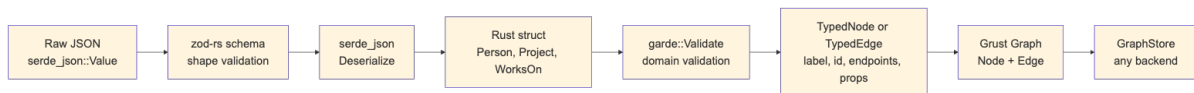


Figure 2: Diagram 2

The important promise is that typed ingestion does not create a second graph model. Once validation succeeds, everything becomes the ordinary Grust graph shape. Backends do not need to know whether a node came from `GraphBuilder`, a YAML document, a typed Rust struct, or a `zod-rs`-validated JSON payload.

garde: Typed Rust Validation

`garde` is a Rust validation library built around a derive macro. You attach validation rules to fields, derive `garde::Validate`, and then call `validate()` or `validate_with(...)`. The rules live near the Rust type, so they travel with the domain model instead of being hidden in a loader function.

Common `garde` rules include:

- `length(min = 1)` for nonempty strings and collections.
- `range(min = 1, max = 100)` for numeric bounds.
- `inner(...)` for validating items inside a collection.
- `custom(...)` for domain-specific validation functions.
- `dive` for validating nested values that also implement `Validate`.

In Grust, a typed node implements `TypedNode`. It supplies a graph label and a stable node id. The default property conversion serializes the struct through Serde and converts the resulting object fields into Grust properties:

```

use grust::prelude::*;
use grust::typed::garde;
use serde::Serialize;

#[derive(Debug, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct Person {
    #[garde(length(min = 1))]
    id: String,
    #[garde(length(min = 1))]
    name: String,
    #[garde(length(min = 1), inner(length(min = 1)))]
    skills: Vec<String>,
}

impl TypedNode for Person {
    const LABEL: &'static str = "Person";

    fn node_id(&self) -> NodeId {
        format!("person:{}", self.id).into()
    }
}

```

Typed edges implement TypedEdge. They provide a label and endpoint ids:

```

#[derive(Debug, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct WorksOn {
    #[garde(length(min = 1))]
    person_id: String,
    #[garde(length(min = 1))]
    project_id: String,
    #[garde(range(min = 1, max = 100))]
    allocation_percent: u8,
}

impl TypedEdge for WorksOn {
    const LABEL: &'static str = "WORKS_ON";

    fn source_node_id(&self) -> NodeId {
        format!("person:{}", self.person_id).into()
    }

    fn target_node_id(&self) -> NodeId {
        format!("project:{}", self.project_id).into()
    }
}

```

TypedGraphBuilder validates and lowers these values:

```

let mut builder = TypedGraphBuilder::new();

builder.add_node(&Person {
    id: "nia".to_string(),
    name: "Nia".to_string(),
    skills: vec!["rust".to_string(), "graphs".to_string()],
})?;

```

```
builder.add_edge(&WorksOn {
  person_id: "nia".to_string(),
  project_id: "grust".to_string(),
  allocation_percent: 80,
})?;
```

```
let graph = builder.build();
```

Validation fails before graph construction. If `allocation_percent` is 0, the `range(min = 1, max = 100)` rule produces a Grust schema error and the edge is not added. This keeps invalid domain facts out of the graph rather than relying on a backend to reject them later.

Typed values can also be reconstructed from ordinary Grust reads. The read path decodes node or edge properties back through `serde`, runs `garde` validation, and checks that the typed identity still matches the graph value:

```
let stored = store
  .get_node(&NodeId::new("person:nia"))
  .await?
  .expect("person exists");
```

```
let person = Person::from_node(&stored)?;
assert_eq!(person.id, "nia");
```

Edges follow the same pattern with `WorksOn::from_edge(&edge)?`. For validation contexts, use `from_node_with` or `from_edge_with`.

Typed and Untyped Graphs Coexist

The typed layer is an ingestion layer over the ordinary `GraphBuilder`. It does not force an all-or-nothing choice. A graph can start as a document, accept raw nodes and edges, and then be extended with typed values:

```
let existing = Graph::new(
  vec![Node::new("Document", "doc:garde-proposal", Props::new())],
  Vec::new(),
);
```

```
let mut builder = TypedGraphBuilder::from_graph(existing);
```

```
builder.add_node(&Person {
  id: "nia".to_string(),
  name: "Nia".to_string(),
  skills: vec!["rust".to_string(), "graphs".to_string()],
})?;
```

```
builder.add_raw_edge(Edge::new(
  "AUTHORED",
  "person:nia",
  "doc:garde-proposal",
  Props::new(),
));
```

```
let graph = builder.build();
```

The coexistence API is explicit:

- `TypedGraphBuilder::from_graph(graph)` starts from an existing `Graph`.

- `TypedGraphBuilder::from_builder(builder)` starts from an existing `GraphBuilder`.
- `add_raw_node(node)` and `add_raw_edge(edge)` add ordinary Grust values.
- `into_builder()` returns the inner `GraphBuilder` when lower-level builder operations are needed.

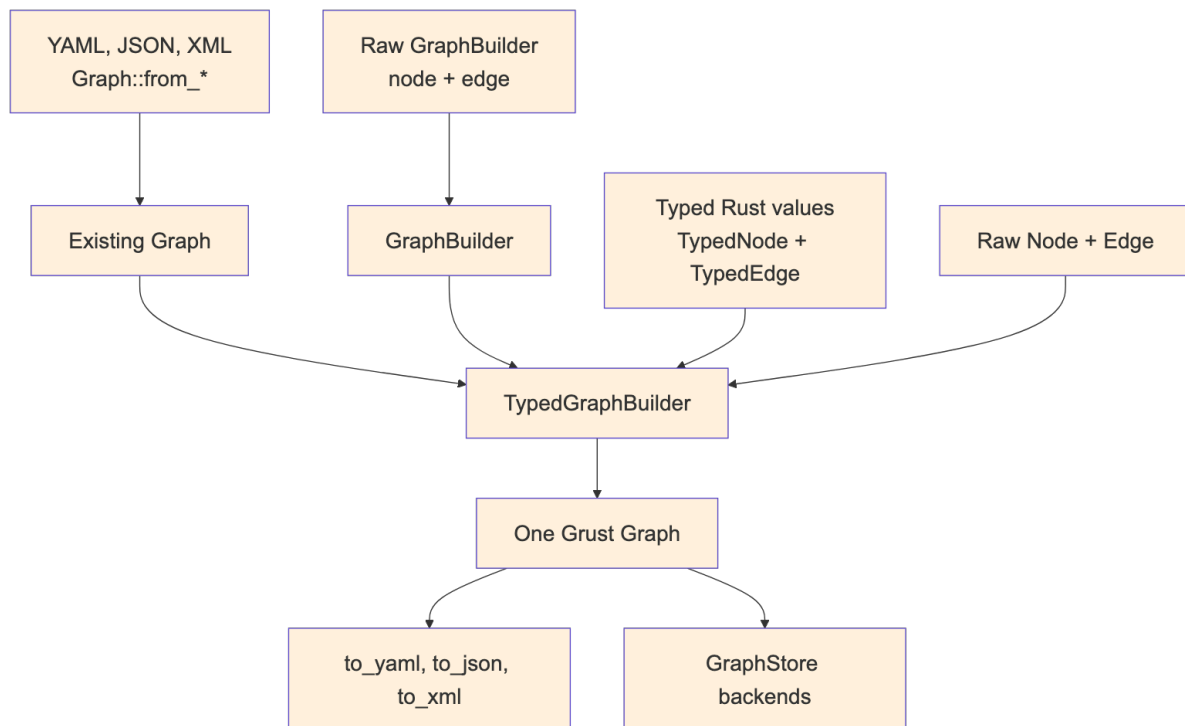


Figure 3: Diagram 3

This is useful during migrations. A project can keep loading existing graph documents while adding typed definitions for the domain facts that benefit most from Rust validation. Over time, more labels can move to typed constructors without breaking the storage or document formats.

What Zod Means Here

Zod is best known from TypeScript. It lets developers define a runtime schema and validate untrusted input before treating it as application data. In TypeScript, this closes an important gap: static types disappear at runtime, so a value received from JSON, an HTTP request, a form, or a message queue still needs runtime validation.

Rust has a different type system, but the boundary problem still exists. External data arrives as bytes or JSON. Serde can deserialize those bytes into a Rust struct, but it is often useful to separate two questions:

1. Does this JSON have the expected shape?
2. Does the resulting Rust value satisfy my domain rules?

`zod-rs` answers the first question. `garde` answers the second. Grust's `typed-zod-rs` feature wires those stages together:

```

use grust::prelude::*;
use serde::{Deserialize, Serialize};
use serde_json::json;
use zod_rs::prelude::{Schema, number, object, string};

#[derive(Debug, Deserialize, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]

```

```

struct Person {
    #[garde(length(min = 1))]
    id: String,
    #[garde(length(min = 1))]
    name: String,
    #[garde(length(min = 1), inner(length(min = 1)))]
    skills: Vec<String>,
}

impl TypedNode for Person {
    const LABEL: &'static str = "Person";

    fn node_id(&self) -> NodeId {
        format!("person:{}", self.id).into()
    }
}

let person_schema = object()
    .field("id", string().min(1))
    .field("name", string().min(1))
    .field("skills", string().min(1).array())
    .strict();

let json = json!({
    "id": "nia",
    "name": "Nia",
    "skills": ["rust", "graphs"]
});

let person: Person = parse_typed_json(&person_schema, &json)?;

```

The same schema can feed the graph builder directly:

```

let mut builder = TypedGraphBuilder::new();
builder.add_node_from_json::<Person, _>(&person_schema, &json)?;

```

For edges, the pattern is the same:

```

#[derive(Debug, Deserialize, Serialize, garde::Validate)]
#[garde(allow_unvalidated)]
struct WorksOn {
    #[garde(length(min = 1))]
    person_id: String,
    #[garde(length(min = 1))]
    project_id: String,
    #[garde(range(min = 1, max = 100))]
    allocation_percent: u8,
}

impl TypedEdge for WorksOn {
    const LABEL: &'static str = "WORKS_ON";

    fn source_node_id(&self) -> NodeId {
        format!("person:{}", self.person_id).into()
    }

    fn target_node_id(&self) -> NodeId {
        format!("project:{}", self.project_id).into()
    }
}

```

```

    }
}

let works_on_schema = object()
  .field("person_id", string().min(1))
  .field("project_id", string().min(1))
  .field("allocation_percent", number().int().min(1.0).max(100.0))
  .strict();

builder.add_edge_from_json::<WorksOn, _>(
  &works_on_schema,
  &json!({
    "person_id": "nia",
    "project_id": "grust",
    "allocation_percent": 80
  }),
)?;

```

The adapter intentionally deserializes the original JSON after zod-rs accepts it. This preserves Rust integer types. For example, zod-rs validates numbers through a floating-point schema, but Serde should still be allowed to decode the original JSON integer 80 into a Rust u8.

How the Options Interact

There are four common construction paths:

Raw GraphBuilder:

trusted Rust code -> Node/Edge -> Graph

Graph documents:

YAML/JSON/XML -> Graph::from_* validation -> Graph

typed-garde:

Rust struct -> garde validation -> TypedNode/TypedEdge lowering -> Graph

typed-zod-rs:

raw JSON -> zod-rs shape validation -> Serde -> garde validation -> Graph

Choose GraphBuilder when the data is already trusted Rust code and the dynamic graph model is enough. Choose graph documents when you need readable fixtures, interchange files, or migration inputs. Choose typed-garde when the domain model lives in Rust structs and should enforce Rust-level invariants. Choose typed-zod-rs when the input is untrusted JSON and you want a separate shape gate before Serde and garde.

These options compose. A single graph can contain nodes loaded from YAML, raw edges added by GraphBuilder, typed nodes validated by garde, and request payloads validated first by zod-rs. The result is still just Graph.

The distinction between zod-rs and garde is worth keeping crisp:

- zod-rs sees `serde_json::Value`.
- Serde converts JSON into Rust types.
- garde sees Rust fields with Rust types.
- TypedNode and TypedEdge convert validated Rust values into Grust labels, ids, endpoints, and properties.

For example, this JSON fails at the zod-rs stage because `skills` is not an array:

```
{
  "id": "nia",
  "name": "Nia",
  "skills": "rust"
}
```

This JSON can pass a loose zod-rs shape check but fail at the garde stage if the Rust type requires at least two skills:

```
{
  "id": "nia",
  "name": "Nia",
  "skills": ["rust"]
}
```

That separation gives application authors precise error boundaries. Shape errors usually belong near the API or file-ingest boundary. Domain errors belong near the typed model.

Relationship to GraphSchema

GraphSchema and typed ingestion solve different problems. GraphSchema describes graph labels, fields, uniqueness, and backend-facing metadata. It can drive indexes, migrations, table layouts, or validation inside a store. Typed ingestion validates values before they become graph data.

In practice they can reinforce each other:

- TypedNode and TypedEdge keep application construction honest.
- zod-rs schemas protect JSON ingress points.
- GraphSchema tells storage backends what structure to expect.
- GraphStore remains the common persistence contract.

This layered design keeps Grust from becoming either too loose or too rigid. Projects can start with raw graph construction, add typed Rust validation for important labels, add zod-rs for JSON ingress, and later add GraphSchema for backend optimization.

The same schema can also validate and write in one step:

```
let report = store.put_typed_graph(&schema, &graph).await?;
```

That call validates labels, required fields, field value types, and edge endpoint labels before delegating to the backend. Schema-capable stores then use `apply_schema` to lower the portable model into their native storage surfaces.

5. Traversal as an Intermediate Representation

Grust traversal is not Cypher, SQL, GQL, SurrealQL, Spark SQL, or a graph database dialect. It is a small Rust intermediate representation:

```
let traversal = Traversal::from_node("talk:rust-graph-api")
  .out("PRESENTED_BY")
  .to("Person")
  .limit(10);
```

A traversal has a start expression, a sequence of steps, and an optional limit. The start can be a single node, all nodes with a label, or nodes with a property value. Each step has a direction, an optional edge label, and an optional target node label.

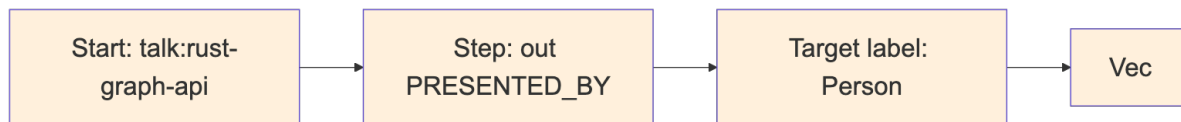


Figure 4: Diagram 4

The IR is deliberately modest. That makes it implementable across a wide range of backends. The memory backend can scan maps. LanceDB can query tables by hop. pgGraph can lower to SQL over universal graph tables. Sail can lower to Spark SQL through Spark Connect. Backends that do not yet support reads can return `GrustError::Unsupported`.

This design also protects application code. A caller asks for a graph-shaped operation; the backend decides whether that operation becomes a map scan, a SQL join, a DataFrame query, a Redis graph command, or a future native graph query.

For local analytics and backend planning, `grust-core` also exposes `GraphIndex`. It builds a dense vertex index from a `Graph`, validates that every edge endpoint exists, and stores incoming and outgoing edge indexes per vertex. That gives adapters and examples one shared adjacency layer instead of rebuilding node-id maps in each backend. The `grust-graph` facade includes a dependency-free benchmarks example that exercises graph cloning, `GraphIndex` construction, degree scans, endpoint scans, and structural edge-key generation over ring, grid, layered, clustered, Graph500-style R-MAT, and GAP-style R-MAT graph families.

6. The Store Contract

The central backend trait is `GraphStore`:

```

#[async_trait::async_trait]
pub trait GraphStore: Send + Sync {
    async fn apply_schema(&self, schema: &GraphSchema) -> Result<()>;
    async fn put_node(&self, node: &Node) -> Result<PutOutcome>;
    async fn put_edge(&self, edge: &Edge) -> Result<PutOutcome>;
    async fn put_graph(&self, graph: &Graph) -> Result<LoadReport>;
    async fn put_typed_graph(&self, schema: &GraphSchema, graph: &Graph) ->
Result<LoadReport>;
    async fn get_node(&self, id: &NodeId) -> Result<Option<Node>>;
    async fn get_nodes(&self, ids: &[NodeId]) -> Result<Vec<Node>>;
    async fn get_edges(&self, query: EdgeQuery) -> Result<Vec<Edge>>;
    async fn traverse(&self, traversal: Traversal) -> Result<Vec<Node>>;
}
  
```

The trait is `async` because real graph stores usually cross a process, network, database, or object-store boundary. The `async_trait` crate hides the current limitations around `async` functions in object-safe traits and lets backend implementations present one uniform interface.

`put_graph` takes `&Graph` instead of consuming `Graph`. That keeps retry, audit, comparison, and multi-backend loading workflows straightforward:

```

let report = store.put_graph(&graph).await?;
backup_store.put_graph(&graph).await?;
println!("loaded {} nodes and {} edges", report.nodes, report.edges);
  
```

Single-element writes return `PutOutcome`. Memory and builder paths can report precise inserted/updated/deduped outcomes. Remote upsert-oriented backends commonly return `Upserted` because distinguishing insert from update would require an extra read or a backend-specific write primitive. Portable callers should treat all written outcomes as success rather than depending on inserted-

versus-updated. LoadReport counts elements that were written or upserted, not necessarily newly created rows.

Administrative operations are split into GraphAdminStore:

```
#[async_trait::async_trait]
pub trait GraphAdminStore: GraphStore {
    async fn bootstrap(&self) -> Result<()> { Ok(()) }
    async fn clear(&self) -> Result<()>;
}
```

This separation keeps ordinary application persistence apart from setup and destructive workflows. A production service may receive a GraphStore, while a test harness or migration tool can require GraphAdminStore.

7. Rust Concepts Used

Grust is a good example of idiomatic Rust applied to a storage abstraction.

Newtypes give semantic weight to strings. NodeId, EdgeId, and Label are cheap wrappers, but they prevent accidental parameter swaps and produce clearer APIs.

Trait-based polymorphism defines the backend boundary. Application code can be generic over impl GraphStore, while each backend owns its own connection, configuration, batching, serialization, and query strategy.

Feature flags keep the facade crate light. The public grust crate re-exports backend crates only when features such as memory, lancedb, pggraph, sail, falkor, helix, surreal, or cocoindex are enabled.

Serde makes the graph model portable. Core types derive Serialize and Deserialize, and backends can turn properties into JSON strings, JSONB, Arrow string columns, or target-state maps.

Interior mutability appears where it is appropriate. The memory backend stores its graph behind Arc<RwLock<...>>, making cloned stores share state while keeping mutation synchronized.

Error typing is explicit. GrustError distinguishes backend, schema, unsupported-feature, and serialization failures. That lets a caller tell the difference between “the database rejected this” and “this backend does not yet support traversal.”

8. Backend Architecture

Backends share the same input model but not the same execution model:

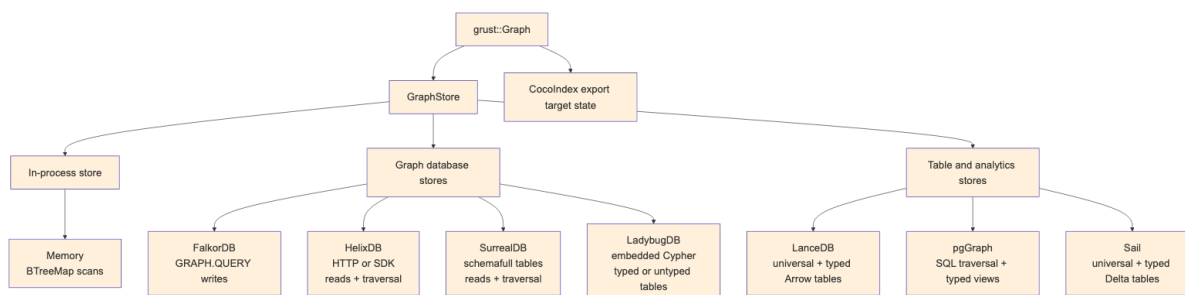


Figure 5: Diagram 5

Some backends are full read/write/traversal stores today. Others still focus on writes and administrative loading. That is normal for an early multi-backend project, and the trait makes the maturity boundary explicit.

Memory

grust-memory is the deterministic local backend. It stores nodes in a BTreeMap<NodeId, Node> and stores edges by source, label, destination, and optional explicit edge ID. That means structural edges still behave deterministically, while id-bearing parallel edges between the same endpoints can coexist. Reads and traversals scan those maps. It is the best backend for tests, examples, and local workflows that need no external service. When a GraphSchema is applied, the memory backend validates writes against it, which makes it a useful conformance harness for typed storage behavior before a database enters the picture.

```
use grust::prelude::*;

# async fn demo(graph: Graph) -> grust::Result<()> {
let store = MemoryGraphStore::new();
store.put_graph(&graph).await?;

let people = store
    .traverse(
        Traversal::from_node("talk:rust-graph-api")
            .out("PRESENTED_BY")
            .to("Person"),
    )
    .await?;
# Ok(())
# }
```

LanceDB

grust-lancedb treats LanceDB first as a durable Arrow-native table store. It uses two tables, one for nodes and one for edges. Nodes are keyed by id. Edges are keyed by an explicit edge id when present, or by a deterministic from + label + to key otherwise. That fallback key is shared with other tabular/export backends through grust-core::edge_key, so structural edge identity does not drift across implementations. Writes use LanceDB merge_insert, and traversal performs hop-by-hop table queries. When a hop fans out to multiple target nodes, the backend reads those target nodes with get_nodes instead of issuing one node query per edge. Property-start traversal filters by label in LanceDB, then compares decoded Grust properties exactly so nested JSON or serialized fragments cannot produce false positives.

This backend is a natural home for later vector-search extensions. The core GraphStore trait should stay graph-focused; LanceDB-specific nearest-neighbor search can live in an extension trait without leaking into every backend.

With GraphSchema, LanceDB also creates typed Arrow tables per node and edge label. The universal grust_nodes and grust_edges tables remain the portable read/traversal surface, while schema-labeled rows are mirrored into tables such as grust_node_person or grust_edge_presents with typed columns for declared fields. That gives analytical consumers and future vector extensions a native columnar surface without giving up the backend-neutral graph model.

LadybugDB

grust-ladybug embeds LadybugDB directly through the Rust lbug crate. It is the durable local graph-database backend: no Docker service, no HTTP bridge, and no separate daemon. The store opens either an in-memory Ladybug database or an on-disk Ladybug directory and creates Grust-managed Ladybug node and relationship tables from graph labels.

Ladybug can be used for typed or untyped property graphs, and the Grust backend now exposes that distinction directly. The default `LadybugGraphMode::Untyped` accepts ordinary Grust graphs and creates the needed Ladybug node and relationship tables from graph labels and endpoint labels on write. That mode matches Grust's raw Graph, JSON, YAML, and XML loading path, where labels and properties are data owned by the application.

`LadybugGraphMode::Typed` requires `apply_schema` or `put_typed_graph` before writes. In that mode the backend creates the declared Ladybug tables up front, does not create undeclared tables during writes, and validates later writes against the applied `GraphSchema`. Reads reconstruct Grust Node and Edge values from the managed tables, while traversal evaluates the portable Grust traversal IR by walking Ladybug relationship tables and reading target nodes through the same `GraphStore` contract.

The `ladybug-arrow` facade feature also exposes Ladybug's embedded Arrow table path through Arrow IPC streams. A caller can register IPC node tables, relationship tables, and CSR relationship tables directly with Ladybug, then query them with Ladybug Cypher and receive result chunks back as Arrow IPC. The public boundary is IPC bytes rather than a Rust `RecordBatch` type, so callers do not have to match Ladybug's internal Arrow crate version exactly.

The first implementation stores Grust properties as JSON text for portable round trips. Later schema lowering can add typed Ladybug columns, full-text indexes, vector indexes, and direct graph-RAG extension traits without changing the core graph model.

pgGraph

`grust-pggraph` stores the source of truth in ordinary PostgreSQL tables:

```
grust_nodes(id text primary key, label text not null, props jsonb not null)
grust_edges(id text, from_id text not null, to_id text not null,
            label text not null, props jsonb not null)
```

The backend can bootstrap the graph extension, create tables and indexes, register graph tables, and optionally build the `pgGraph` projection. Reads use SQL against the universal tables. Traversal is lowered to SQL joins over those tables, with `pgGraph` projection support available as the backend matures. Mutation batches are wrapped in PostgreSQL transactions.

Schema application adds typed label views and expression indexes. For example, a Person node schema with `name: String` and `age: Int` can produce a `grust_node_person` view over the universal node table, with `age` exposed as a `bigint` expression. This is a deliberately incremental typed-storage path: PostgreSQL keeps the flexible JSONB source of truth while callers that know the schema get typed SQL surfaces.

Sail

`grust-sail` connects to a Sail Spark Connect server over gRPC. It stores graph data in Spark DataFrames backed by Delta tables. SQL commands and reads are sent as Spark Connect SQL relation plans, and read results are decoded from Arrow IPC streams.

Bulk writes stage Arrow IPC batches as Spark Connect `LocalRelation` temp views and then merge from those views. That avoids building one giant SQL literal per row, keeps user values out of SQL text, and gives long-running requests an operation id with reattachment enabled. Query filters bind user values through Spark Connect named arguments. Delete mutations stage their values as Arrow temp views before running argument-free SQL commands, which avoids string substitution while matching Sail's current command-parameter behavior. Traversal joins use globally unique node ids;

source and destination label columns may be empty for single-edge writes where the full graph is not in scope.

Sail's Arrow boundary is now public too. Applications can stage arbitrary Arrow IPC streams as session temp views and query them with Spark SQL, collect Spark SQL results as Arrow IPC chunks, or load Grust-shaped node and edge IPC streams through the normal graph write path. This gives Sail the same data-source role as Ladybug while keeping Sail's internal Arrow 58 dependency separate from Ladybug's Arrow 55 dependency.

When a schema is applied, Sail creates typed Delta tables per node and edge label and mirrors writes into them with `MERGE INTO`. The universal Spark tables keep traversal simple and portable; the typed tables make declared graph labels available as ordinary Spark columns.

Sail also has reusable graph analytics helpers over the persisted generic tables. `read_graph` collects the generic `grust_nodes` and `grust_edges` tables back into a portable Grust Graph. `in_degrees`, `out_degrees`, `degrees`, and `degree_pairs` run Spark SQL over those same tables and decode the Arrow results into small Rust row types. These helpers are deliberately low-level: they expose common graph measurements without making the backend-neutral `GraphStore` trait depend on Spark-specific analytics.

The Sail crate now also publishes its generic table contract. Constants name `grust_nodes`, `grust_edges`, and the physical node and edge columns, including the persisted generic edge `edge_key` and optional explicit edge `id`. Projection helpers classify requested graph fields as physical columns or JSON properties, map edge label to the stored `edge_type`, and check when a typed Sail node or edge table can satisfy a graph query directly. This keeps Sail-native graph planning and GrustFrames-style distributed lowerings aligned with the same backend layout that Grust writes.

The same contract layer exposes typed table descriptors derived from `GraphSchema` and directional triplet SQL. The triplet helpers join generic edges back to their source and destination nodes, and can orient rows as outgoing, incoming, or undirected pairs. That is the shared primitive needed by distributed triplet filters, motif expansion, and aggregate-message style passes without making the backend-neutral store trait depend on those higher level algorithms.

Writable Cypher in Sail follows the same rule. `sail_cypher_mutation_plan` accepts only a strict v1 mutation subset: explicit-ID node `CREATE` and `MERGE`, edge `CREATE` and `MERGE` when both endpoint IDs are resolved, and resolved node or edge `DELETE`. It also accepts ordered multi-statement batches, local node variables bound by explicit-ID node patterns, ID-resolved `MATCH ... DELETE`, edge `MATCH ... CREATE / MATCH ... MERGE`, and cardinality-aware broad node `MATCH ... DELETE / MATCH ... SET n += { ... } / MATCH ... SET n.key = value / MATCH ... REMOVE n.key` forms, plus ID-resolved edge `MATCH ... SET e += { ... }`, literal property assignment, and explicit property `REMOVE` when the node or edge identity is resolved, row-producing edge `MATCH ... CREATE` and `MATCH ... MERGE` when both endpoints come from matched node variables, plus broad relationship `MATCH ... DELETE / MATCH ... SET e += { ... } / MATCH ... SET e.key = value / MATCH ... REMOVE e.key` over endpoint predicates. It lowers those statements into `GraphMutationPlan` and then ordinary `GraphMutation` values. The execution entrypoint, `SailGraphStore::execute_cypher_mutation`, runs those mutations through `GraphMutationStore`, so Cypher writes use the same staged Arrow, `MERGE INTO`, typed-table mirror, and delete paths as normal Grust writes. When callers need stricter Cypher compatibility, the options entrypoint can make `CREATE` fail if the target node or edge identity already exists. That mode performs a read-before-write check, rejects duplicate concrete node or edge `CREATE` identities inside one planned batch before any writes run, and therefore keeps the default entrypoint on the lower-friction upsert-compatible path. Generated node IDs are also caller-selected rather than implicit:

CypherNodeIdPolicy::GenerateForCreate allows node CREATE without an id, and the result API returns generated IDs separately from the count-oriented mutation report. Callers can also opt into written node and edge identity payloads. Node payloads cover explicit and generated node writes; edge payloads cover resolved and row-producing edge writes. These payloads describe accepted writes rather than exact insert-versus-update outcomes on upsert backends. MERGE and edge endpoint patterns still require resolved IDs before writing. The first table-returning write path is deliberately smaller than general Cypher RETURN:

execute_cypher_mutation_returning_with_options accepts a final projection over node variables and relationship variables already resolved by the write plan, including concrete edge upserts and edge patches. Sail and the backend-neutral Memory/Sail helper can also return one row per relationship variable produced by restricted row-producing MATCH ... CREATE/MERGE edge writes, plus portable broad node rows for restricted MATCH ... SET/REMOVE forms such as MATCH (n:Person {status: 'active'}) SET n.seen = true RETURN n.id, n.seen and portable broad relationship rows for restricted MATCH ... SET/REMOVE forms such as MATCH (a)-[e:KNOWS]->(b) SET e.seen = true RETURN e.id, e.seen. Physical id and label fields are supported alongside stored properties, for example RETURN n.id, n.label, n.seen, RETURN e.id, e.label, e.weight, or RETURN e.label, e.source after a row-producing edge write; whole elements can also be returned as Value::Json in the Grust Node / Edge serde shape with RETURN n AS node, e AS relationship. Restricted Cypher function spelling is also available over those already-bound values, including labels(n), type(e), properties(n), keys(e), id(n), elementId(e), exists(n.nickname), size(n.nickname), toLower(n.team), isEmpty(n.nickname), toString(n.score), toUpper(n.code), abs(n.score), ceil(n.score), floor(n.score), sign(n.score), toInteger(n.score), toFloat(n.score), trim(n.name), substring(n.name, 0, 3), replace(n.team, '-team', ''), startsWith(n.team, 'eng'), left(n.team, 3), right(n.code, 2), reverse(n.code), split(n.path, '/'), startNode(e), and endNode(e). Relationship identity functions return an explicit relationship ID when one exists and null otherwise; they do not invent generated structural identity. The returning endpoint returns CypherMutationTableResult, which keeps mutation reporting separate from CypherResultTable. General expression evaluation, arbitrary read-query features, unrestricted broad row materialization, and variable-length path semantics remain rejected until a shared read/write row model owns those semantics.

CypherMutationOptions::parameters binds Grust Values to \$name placeholders only where literals are already accepted: IDs, property maps, and literal property assignments. Quoted '\$name' remains ordinary string text rather than a parameter reference. Mutating MATCH clauses can use a bounded WHERE grammar: property comparisons against literals or parameters joined by AND, for example WHERE n.status = 'inactive' AND n.score >= \$min. Predicates lower to backend-neutral GraphPropertyPredicate values, so Memory evaluates the same resolved plan that Sail lowers to SQL. Missing properties never match; null only matches equality or inequality against Value::Null, and ordered comparisons are limited to numbers or strings.

CypherMutationOptions::null_assignment defaults to storing SET x.key = null as Value::Null, but callers can choose CypherNullAssignment::RemoveProperty to lower explicit null assignment to the same property-removal operations used by REMOVE; map patches such as SET x += {key: null} always store Value::Null. MATCH ... SET clauses can contain comma-separated assignments, and each assignment is lowered as its own ordered plan operation so repeated property targets preserve source order while still using only the supported literal, map patch, remove-on-null, and numeric node update forms. The first expression form is deliberately small: node property assignment can read a property on the same node variable and apply +, -, *, or / with an integer or float literal or parameter. That lowers to an explicit read-modify-write mutation plan instead of hidden parser-side arithmetic. Resolved mutation-plan execution is backend-neutral through CypherMutationExecutor: Sail still owns text parsing, but the resulting GraphMutationPlan can

execute on Sail or on the in-memory backend for deterministic tests. Backends that cannot execute a plan operation report a structured Cypher execution error. The Sail parser has an internal front-door boundary that classifies top-level mutation statements before lowering; a shared parser crate remains deferred until there is a second Cypher text parser consumer. Mutation batch atomicity is explicit through `GraphMutationAtomicity`: the default path is ordered but not atomic, while `pgGraph` and `SurrealDB` report transactional batch execution because they wrap mutation batches in backend transactions. ID-resolved and broad node `MATCH ... SET n += { ... }` lower to node patch mutations; `null` in the patch map is stored as `Value::Null` rather than treated as property removal. ID-resolved edge `MATCH ... SET e += { ... }` lowers to an edge patch mutation and reuses the same typed-edge mirror writes as ordinary edge upserts. Row-producing edge `MATCH ... CREATE` and `MATCH ... MERGE` materialize matched endpoint node pairs before writing, report matched rows separately from attempted edge upserts, and still reject trailing node creation or explicit relationship IDs in the row-producing form. The current report does not distinguish newly inserted merge rows from rows that already existed. Literal `SET n.key = value` and `SET e.key = value` lower to one-key patches, while `REMOVE n.key` and `REMOVE e.key` lower to explicit property-remove mutations. Node forms can target either a resolved identity or a broad node match; edge forms can target either a resolved identity or a broad relationship match. Broad relationship matches can filter on relationship property predicates beyond `id`; explicit edge `id` remains a separate identity filter and can be combined with ordinary relationship predicates. Relationship expression updates and general computed expressions remain deferred. For broad node and relationship deletes, patches, and property removals, the report records matched rows and changed graph elements, and Sail stages matched IDs before using the same delete or load helpers that keep generic and typed tables consistent. The mutation parser keeps top-level keywords case-insensitive and strips Cypher comments outside string literals. Callers can distinguish Cypher syntax, unresolved identity, unsupported cardinality, and execution failures through structured `GrustError` variants. Execution remains Sail-specific for now, while the mutation plan and report types stay backend-neutral.

FalkorDB, HelixDB, and SurrealDB

The FalkorDB backend writes through Redis `GRAPH.QUERY` using Cypher-like `MERGE` statements. It batches nodes by label path and edges by relationship type. Schema application creates label/property indexes for declared node types.

The HelixDB backend has HTTP and SDK stores. Both support batched writes, node reads, edge reads, and backend-neutral traversal through Helix dynamic queries. Edge writes store Grust relationship metadata (`relationship`, `from_id`, `to_id`, and optional `edge_id`) so `EdgeQuery` can reconstruct Grust edges from Helix relationship rows. Helix writes preserve supported scalar and array properties instead of silently dropping non-string values; unsupported JSON object properties return an explicit error. The current schema hook validates that labels, relationships, and fields can be safely lowered through the dynamic-query path; backend-native schema-file generation can build on that same `GraphSchema` contract later.

The SurrealDB backend also has HTTP and SDK stores. It can bootstrap, clear, upsert nodes, relate edges, delete nodes and edges, read nodes and edges, and execute traversal hop-by-hop through the `GraphStore` contract. Applying a schema lowers node and edge declarations to Surreal `DEFINE TABLE` and `DEFINE FIELD` statements so the backend can run in `schemafull` mode where the schema calls for it. Reads use configured labels and relationships, plus ID-derived table names, to find records across Surreal's label-specific tables. Generic edge reads and node deletes require `SurrealConfig.relationships`; when that list is empty, the backend returns a configuration error instead of silently scanning no relation tables. Explicit edge-label reads and deletes can still target a known relation table directly. Traversal batches target-node reads per step through `get_nodes`,

avoiding a serial node lookup for every edge in a fan-out. Mutation batches are wrapped in SurrealDB transactions.

CocoIndex

grust-cocoindex is intentionally not an ordinary GraphStore. CocoIndex is an incremental target-state system, so the Grust integration exports a graph as serializable node and relationship state:

```
let export = graph.to_cocoindex_export()?;
let graph = cocoindex_export_to_graph(export)?;
```

The export uses Grust node IDs as target keys, converts properties to JSON, and requires edge endpoints to exist in the graph so relationship source and target labels can be emitted. The import path expects the same key shape, validates relationship endpoints against imported nodes, and preserves explicit relationship keys as Grust edge IDs.

Backend Integration Tests

Unit tests stay self-contained, but live backend tests are explicit. They are marked ignored in Cargo so a normal workspace test run does not accidentally depend on local services. When a live test is requested, however, it must reach the backend; it no longer returns early and pretends success when a server is missing.

The repository provides a launcher for those checks:

```
scripts/integration-test.sh doctor --profile docker --mode docker
scripts/integration-test.sh --profile docker --mode docker
```

The Docker profile is the contributor path: it starts the Docker-backed services and runs local LanceDB and CocoIndex checks. The full maintainer profile is still available:

```
scripts/integration-test.sh --profile all
```

The launcher reads integration/backends.conf. In auto mode it prefers already-running services, then configured local source checkouts such as ~/src/sail, ~/src/SurrealDB, ~/src/FalkorDB, and ~/src/HelixDB, then Docker Compose where a service is available. The repository-level docs/INTEGRATION.md guide covers profiles, modes, Docker image pins, source-checkout configuration, and CI strategy.

9. Example: A Conference Graph

Here is a complete graph-building example:

```
use grust::prelude::*;

fn conference_graph() -> Graph {
    let mut g = Graph::builder();

    let rust = g
        .node("Conference", "conf:rust-graph-day")
        .prop("name", "Rust Graph Day")
        .prop("city", "San Francisco")
        .finish();

    let talk = g
        .node("Talk", "talk:backend-neutral-graphs")
        .prop("title", "Backend-Neutral Graphs in Rust")
        .prop("track", "systems")
        .finish();
```

```

let ada = g
  .node("Person", "person:ada")
  .prop("name", "Ada Example")
  .prop("role", "speaker")
  .finish();

g.edge("HAS_TALK", &rust, &talk).finish();
g.edge("PRESENTED_BY", &talk, &ada)
  .prop("confirmed", true)
  .finish();

g.build()
}

```

The resulting graph can be loaded into memory, exported to CocoIndex target state, or sent to a configured backend:

```

# use grust::prelude::*;
# async fn load<S: GraphStore>(store: &S, graph: Graph) -> grust::Result<()> {
let report = store.put_graph(&graph).await?;
assert_eq!(report.nodes, 3);
assert_eq!(report.edges, 2);
# Ok(())
# }

```

The same graph can be loaded through the typed backend path by declaring the schema and using `put_typed_graph`:

```

# use grust::prelude::*;
# async fn typed_load<S: GraphStore>(store: &S, graph: Graph) -> grust::Result<()> {
let schema = GraphSchema::builder()
  .node(
    "Conference",
    vec![
      Field::required("name", FieldType::String),
      Field::required("city", FieldType::String),
    ],
  )
  .node(
    "Talk",
    vec![
      Field::required("title", FieldType::String),
      Field::required("track", FieldType::String),
    ],
  )
  .node(
    "Person",
    vec![
      Field::required("name", FieldType::String),
      Field::required("role", FieldType::String),
    ],
  )
  .edge(
    "HAS_TALK",
    vec![Label::new("Conference")],
    vec![Label::new("Talk")],
    Vec::<Field>::new(),
  )
}

```

```

    )
    .edge(
        "PRESENTED_BY",
        vec![Label::new("Talk")],
        vec![Label::new("Person")],
        vec![Field::required("confirmed", FieldType::Bool)],
    )
    .build();

let report = store.put_typed_graph(&schema, &graph).await?;
assert_eq!(report.nodes, 3);
assert_eq!(report.edges, 2);
# Ok(())
# }

```

That call checks node labels, edge labels, required fields, field types, and edge endpoint labels before writing. Backends then use the same schema in their own way: memory validates, LanceDB mirrors into typed Arrow tables, Sail mirrors into typed Delta tables, pgGraph exposes typed SQL views and indexes, SurrealDB defines schemafull tables and fields, and FalkorDB creates useful indexes.

Traversing from a conference to speakers becomes a backend-neutral expression:

```

let speakers = store
    .traverse(
        Traversal::from_node("conf:rust-graph-day")
            .out("HAS_TALK")
            .to("Talk")
            .out("PRESENTED_BY")
            .to("Person"),
    )
    .await?;

```

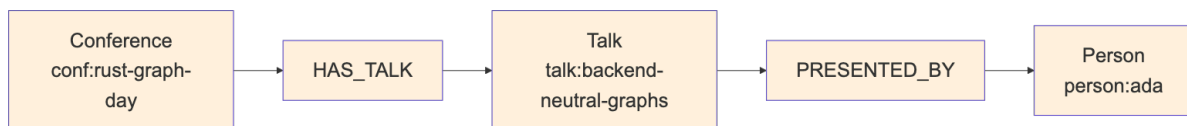


Figure 6: Diagram 6

10. Schema and Validation Direction

Grust has schema types: `GraphSchema`, `NodeType`, `EdgeType`, `Field`, `FieldType`, `EdgeUniqueness`, and `GraphConstraint`. `GraphSchema` validates labels, required fields, field value types, edge endpoint labels, edge direction, declared edge uniqueness, and required-property constraints. `FieldType` includes scalar strings, integers, floats, booleans, RFC 3339 date-times, string/int/float arrays, and JSON. Date-time values use the opaque `RfcDate` type internally; construct them through `Value::datetime` or `RfcDate::parse` so invalid strings cannot bypass validation. The default `GraphStore::apply_schema` implementation remains a no-op, which lets schemaless or schema-later backends work without ceremony.

Schema becomes more important for backends that want typed tables, indexes, or label-partitioned layouts. The current schema-capable backends use it in different ways:

- Memory stores the schema and validates local writes.
- FalkorDB creates label/property indexes for declared node types.
- Helix validates schema names that can be lowered through dynamic queries.

- LadybugDB can run in untyped dynamic mode or typed schema-applied mode; typed mode validates writes against the applied schema.
- LanceDB creates typed Arrow tables per node and edge label and mirrors writes into them.
- pgGraph/PostgreSQL exposes typed label views and expression indexes over the universal tables.
- Sail creates typed Delta tables per node and edge label and mirrors writes into them.
- SurrealDB lowers schema into `DEFINE TABLE` and `DEFINE FIELD` statements.

`apply_schema` is therefore a backend metadata hook, not a portable promise that every future write is enforced by the database. `put_typed_graph` always validates the whole graph with `GraphSchema::validate_graph` before applying schema metadata and writing. Callers that need backend-independent guarantees can run the same validation before ordinary `put_graph` or single-element writes. Memory and LadybugDB enforce the applied schema on subsequent local writes; Sail and LanceDB validate writes before mirroring them into typed tables. FalkorDB, Helix, pgGraph, and SurrealDB currently use schema primarily for indexes, query-shape validation, views, or backend-native definitions. Constraint handling follows the same honest-capability rule. Required-property constraints validate through `GraphSchema`; unique-property constraints validate inside `GraphSchema::validate_graph`, and the memory backend reports validate-before-write behavior for them. Backends that have not added a read-before-write preflight or native enforcement should continue reporting metadata-only behavior through `GraphStore::constraint_capability`.

A flexible backend can keep universal node and edge tables as the portable interchange surface. A typed backend can add native tables, fields, indexes, or constraints behind the same `GraphStore` trait.

The key architectural point is that schema is metadata about a Grust graph, not a replacement for the graph. Application code can begin with plain graph construction, add schemas when operational needs demand it, and still speak the same store trait.

11. Design Tradeoffs

The universal node/edge layout appears in multiple backend plans because it is the easiest way to preserve arbitrary property graphs:

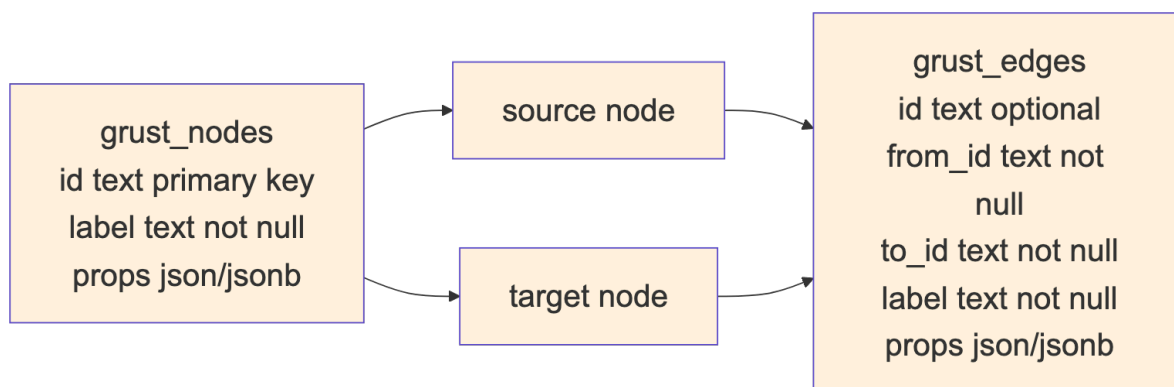


Figure 7: Diagram 7

The cost is that property typing and label-specific optimization require extra work. A label-partitioned backend can use stronger types and better indexes, but it needs schema, migrations, and more planning.

Grust's current architecture keeps both paths open. Core stays universal. Backends choose their storage layout. Schema support can become richer without forcing every backend to look the same.

12. Where Grust Can Grow

The next natural step is to deepen graph-native read and traversal support across the backends. HelixDB, LadybugDB, and SurrealDB now satisfy the portable GraphStore read and traversal surface, while FalkorDB remains primarily a write and indexing adapter. Further work can push more traversal work into backend-native query forms and add richer result shapes.

Traversal can also grow carefully. Property filters, bounded depth, path returns, shortest paths, and aggregation are all tempting. The important rule is to extend the IR only when several backends can implement the concept without smuggling database-specific query strings through the abstraction.

Incremental mutation now has a small extension trait for backends that can delete elements:

```
pub enum GraphMutation {
    UpsertNode(Node),
    DeleteNode(NodeId),
    UpsertEdge(Edge),
    DeleteEdge {
        from: NodeId,
        label: Label,
        to: NodeId,
    },
}
```

That serves CocoIndex-style target-state systems, streaming pipelines, and ordinary applications that need to apply deltas instead of replacing whole graphs. The default `apply_mutations` implementation is ordered but not atomic: if a backend uses the default and a later mutation fails, earlier mutations may already be committed. Backends with real transaction support can override that method. The `pgGraph` backend wraps mutation batches in PostgreSQL transactions, and the SurrealDB HTTP and SDK stores wrap mutation batches in SurrealDB transactions.

Conclusion

Grust is small by design. Its core model is easy to hold in your head, and its backend contract is narrow enough that very different systems can implement it. That is the source of its leverage.

The project says: build your graph once, keep the domain model in Rust, and let the backend translate. Sometimes that translation is a map scan. Sometimes it is Arrow and LanceDB. Sometimes it is PostgreSQL, Spark SQL, Redis graph commands, SurrealQL, Helix SDK calls, or CocoIndex target state.

The more Grust grows, the more important that center becomes: a stable property graph model, a backend-neutral traversal IR, explicit errors, feature-gated integrations, and enough Rust type structure to make the right path feel natural.